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Review papers

Causes and implications of groundwater depletion in India: A review

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ABSTRACT

Groundwater plays a vital role in food security, water availability, and economic growth of more than 1.3 billion people in India. Notwithstanding the development of irrigation schemes based on surface water, groundwater remains a primary source of irrigation in large parts of the country. The Green revolution in the 1970s contributed considerably to increasing food production in India through the development of groundwater-based irrigation, which resulted in an expansion of irrigated area and a rapid rise in the number of electric and diesel pumps. A large part of the rain-fed area was converted to irrigated agriculture leading to growing multiple crops in a year with increased crop yields. While the expansion in groundwater-based irrigation helped meet the rising food demands of a large population of India, it has resulted in several environmental implications. Groundwater depletion in India has now become one of the most prominent challenges for food and water security. Both in-situ and satellite-based observations indicate a rapid decline in groundwater storage in north India. Both natural and anthropogenic factors cause groundwater depletion in India. Groundwater pumping for irrigation remains the primary driver of groundwater depletion, which can further affect food and water security in India under climate change. Both in-situ and satellite-based measurements have issues and uncertainty. Land surface hydrological models provide essential insights in understanding the groundwater storage variability; however, need improvements in the representation of complex processes related to climate-human interactions. Sustainable management of groundwater in India is vital for tackling growing challenges related to water availability. Here we discuss the challenges and opportunities related to the measurements and modelling of groundwater, groundwater recharge, cropping systems and irrigation efficiency, and social and policy reforms for sustainable groundwater management in India.

1. Introduction

Climate change and population growth have significantly affected the global water resources in the 20th century and are likely to pose more challenges in the future (Milly et al., 2005; Vörösmarty et al., 2000). Groundwater is an essential source of freshwater meeting the domestic needs of about half of the global population (Zekt˙s˙er et al., 2004). About 38% of global consumptive irrigation water demand is met by groundwater (Siebert et al., 2010). Depletion of groundwater is significant in most of the irrigated regions in the world where millions of people reside (Kerr, 2009). Groundwater abstraction accounts for 33% of the total withdrawals globally and provides water for agricultural (42%), domestic (36%), and industrial (27%) needs (Döll et al., 2012; Taylor et al., 2013). The exploitation of groundwater is mainly due to the quick expansion of irrigated lands to meet the needs of food demands of the increasing population (Giordano, 2009; Wada et al., 2012). Irrigated water use for food production for more than 50% population is from groundwater (Famiglietti, 2014) in around 39 million ha irrigated land (Siebert et al., 2010). Gleeson et al. (2012b) estimated current global groundwater footprint to be 3.5 times the actual aquifer area threatening about 1.7 billion people, while 80% of aquifers having footprint smaller than their area, implying that only a few overexploited aquifers drive the net global footprint. Groundwater depletion is mainly concentrated in localized overexploited aquifer systems (Wada et al., 2012) as in Northern India (Rodell et al., 2009).

India has the most substantial extent of the irrigated area with groundwater (Siebert et al., 2010) [Fig. 1]. A large part of the country is under intensive agriculture (Fig. 1a) and mostly irrigated by groundwater (Fig. 1b). Irrigation is a significant contributor to increased food production in densely populated India (Fig. 1c). India has 4% of total global land, hosting 24% of the worldwide population, and having greater than 30% of irrigated land in the world (FAO, 2013). Irrigated

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area in India has increased from 58.8% to 60.4% from 1961 to 2016 (World Bank, 2017). The net irrigated area from different sources (canals, tanks, wells, and tube-wells and others) has increased from 56.94 Mha to 68.38 Mha from 2001 to 2015 (MOSPI, 2018). Global groundwater depletion for irrigation has increased from 2000 to 2010, and 23% of which is contributed by India (Dalin et al., 2017). India is an epitome for assessing the impact of energy policy on groundwater as the power needed for water extraction is either free or highly subsidized in the significant regions of the country (Giordano, 2009).

In water-limited areas, irrigated agriculture is more suitable than rainfed agriculture (Cassman, 2016; Mueller et al., 2012). Mismanagement of water resources can lead to more reliance on groundwater even in the rain abundant areas and also due to uneven distribution of rainfall (Rodell et al., 2009). India is mainly dependent on the Indian summer monsoon rainfall (ISMR), and weaker summer monsoon can cause droughts, as in 2009 (Goldin, 2016). During such dry periods, water abstracted from groundwater results in a reduction in groundwater levels and thereby can worsen the drought conditions (Van Loon et al., 2016). Therefore, groundwater is an essential resource for the sustainability of ecosystems and livelihoods in the second most populated country in the world. Also, groundwater can minimize the adverse impacts of climate variability on water availability.

Rapid groundwater depletion remains one of the most profound sustainability challenges that India is facing currently. Sustaining groundwater is essential for meeting the needs of food and water in the current and projected future climate. However, the current status of the monitoring and management of groundwater depletion, and its implications, and knowledge gaps in India have not been systematically reviewed. Here we provide an overview of groundwater depletion in India and the sustainability challenges it poses (Fig. 2). First, we discuss the overview of the spatial and temporal variability of the groundwater storage in India available from in-situ and satellites. Then, we provide an overview of the consequences of groundwater depletion in India. Next, the management strategies for the sustainable usage of groundwater resources in India are discussed. We discuss the challenges and opportunities and the key areas that need enhancements for sustainable

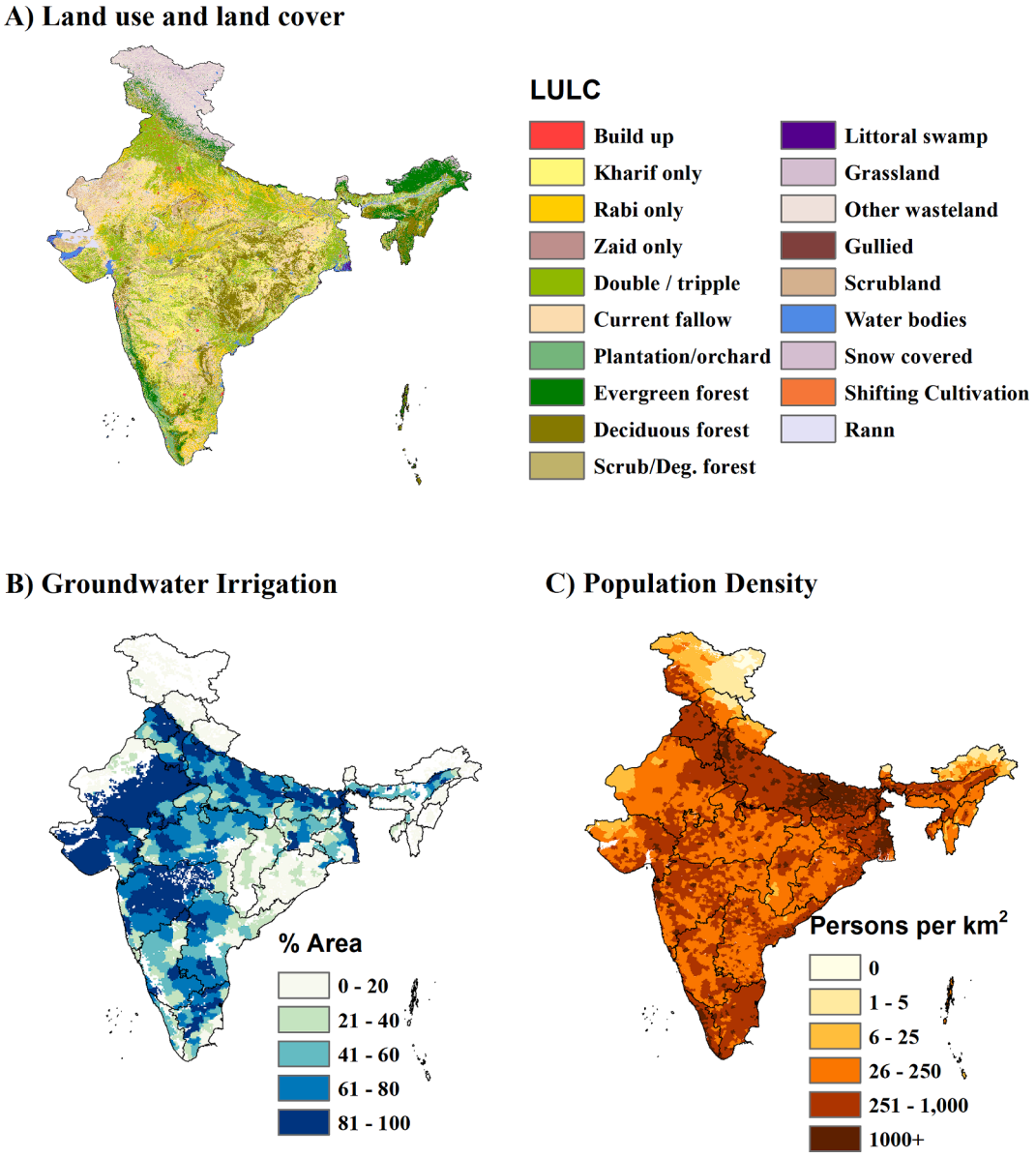


Fig. 1. A) Land use land cover map of India obtained from the National Remote Sensing Center (NRSC) B) The percentage of area irrigated with groundwater from Food and Agricultural Organization (FAO), C) The population density map of India for the year 2015 obtained from the Gridded Population of the World, Version 4 (Center for International Earth Science Information Network - CIESIN, 2018).

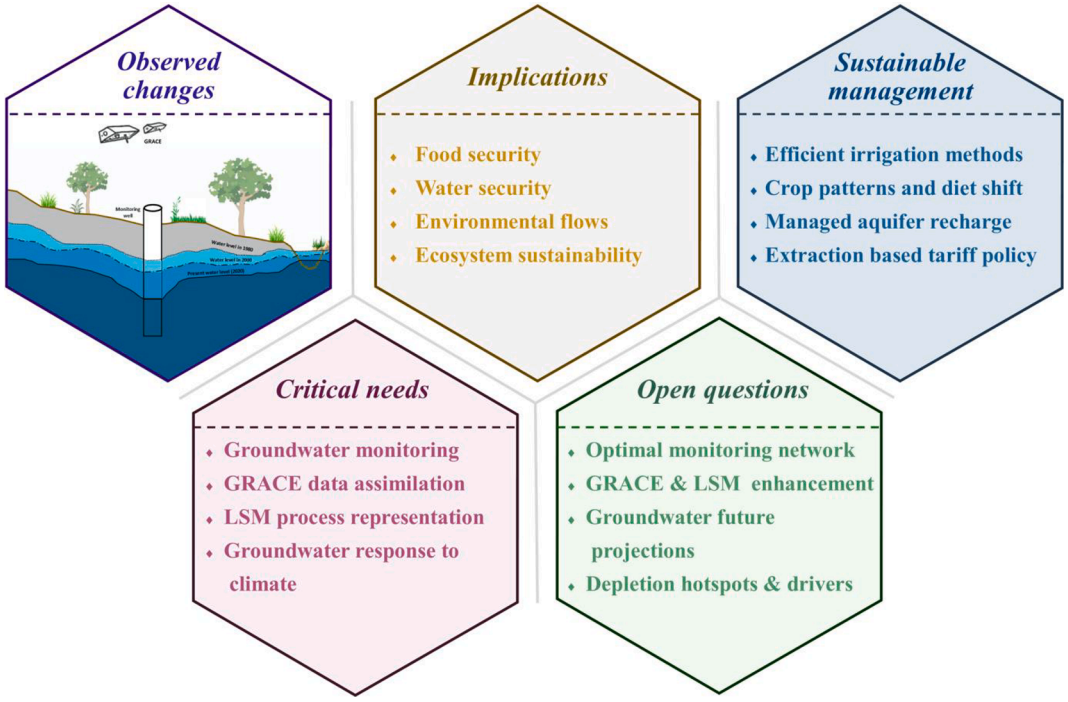


Fig. 2. Overview of key components of groundwater depletion, implications, needs, and open questions in India.

groundwater management in India. Finally, we provide knowledge gaps and open questions related to groundwater management in India.

2. Observed changes in groundwater storage in India

Understanding of the observed changes in groundwater storage is vital for a proper examination of the issues and challenges related to groundwater sustainability in India. Both natural and anthropogenic factors have driven observed changes in groundwater storage. For instance, droughts and floods can lead to groundwater depletion and rise in the water table, respectively. On the other hand, groundwater pumping for irrigation can cause a long-term deficit in groundwater storage in India. Both satellite-based and in-situ operations show variations in the changes in groundwater storage, which warrants a need to discuss the spatial and temporal variability of groundwater storage in India. In this section, we provide a detailed overview of the observed changes in groundwater storage in India.

India has several groundwater stressed aquifers in the world (Gleeson et al., 2012b) [Fig. 1]. Stressed groundwater aquifers are also found in the western part of Ganges-Brahmaputra aquifer systems, whereas the eastern part is unstressed (Fig. 3), consistent with the CGWB estimation (Gleeson and Wada, 2013). This contrast is due to the combination of more recharge rates and less consumption in lower Ganges compared to Upper Ganges aquifer (Gleeson et al., 2012b). These stressed aquifers determine the national average time series of groundwater data. The northwest region lost groundwater storage of around 109 km^3 in six years (2002–2008), which is more than triple the volume of the largest human made reservoir in the USA (Rodell et al., 2009). Northwest India has high groundwater abstraction rate with water table going beyond 20 m and declining at a rate greater than 1 m yr^{-1} (MacDonald et al., 2016; Mishra et al., 2018). Punjab is the hotspot region, representing the highest nonrenewable groundwater extraction of 34.66 km^3 (Panda and Wahr, 2016), whereas renewable volume is only 20.35 km^3 (CGWB, 2014). In Punjab, groundwater abstraction is considerably higher than mean annual rainfall, leading to unsustainable usage (Mishra et al., 2018).

Groundwater depletion in India has a large spatial variability. Groundwater depth during 1996–2013, from most of CGWB observation wells located above 23° north, showed a significant decline ~ 15–25 cm yr^{-1} ($p < 0.05$) (Asoka et al., 2017) [Fig. 3]. Rodell et al. (2018) estimated the terrestrial water storage (TWS) depletion rate in northern India to be 19.2 ± 1.1 Gt yr^{-1} during 2002–2016 as the abstractions are already exceeding the groundwater recharge during normal monsoon years (Mishra et al., 2018). In the north and central India, groundwater is abstracted during the post and pre-monsoon seasons to provide irrigation for rice and wheat crops (Asoka et al., 2017; Panda and Wahr, 2016; Rodell et al., 2018). Meghwal et al., (2019) analyzed the Northwest Indian states of Gujarat and Rajasthan from 2002 to 2016 and found groundwater storage increased in Gujarat by 1.76 km^3 yr^{-1} while declined in Rajasthan by −4.17 km^3 yr^{-1}. A negative trend of 2 cm yr^{-1} ($p < 0.05$) in groundwater anomalies is observed in the majority of north India from 2002 to 2013 (Asoka et al., 2017) and north-east parts of India from the Gravity Recovery and Climate Experiment (GRACE) satellites. Rise in the groundwater table in most parts of south India was observed after 2005 (Tiwari et al., 2011). Asoka et al. (2017) also reported an increase (~1–2 cm yr^{-1}) in GRACE groundwater storage anomalies in south India, consistent with the changes based on the observation wells during the 2002–2013 period. For instance, a significant increase of 5–20 cm per year was observed from wells and GRACE data in south India (Asoka et al., 2017). Bhanja et al. (2017) reported a replenishment rate of 2.04 ± 0.20 km^3 yr^{-1} in 2002–2014 in the western part and 0.76 ± 0.08 km^3 yr^{-1} in 2003–2014 in the southern part of India. Rodell et al. (2018) also reported an increasing trend (9.4 ± 0.6 Gt yr^{-1}) in groundwater storage in Central and South India during 2002–2016. Please see Table 1 for a detailed comparison of changes in groundwater in different studies.

The rising trend in groundwater storage in South India can be attributed to the increase in the monsoon season rainfall (Asoka et al., 2017) and changes associated with electricity subsidies for groundwater extraction (Bhanja et al., 2017; Meghwal et al., 2019; Shah, 2012). The southern and central India show different trends from in-situ well observations that cannot be assessed using the GRACE satellites, primarily due to their coarse spatial resolution (Scanlon et al., 2012). A few opposite trends can also be seen in the north and south India groundwater levels, thus, showing a complex nature of localized groundwater due to pumping requirements and recharge from nearby sources (Asoka

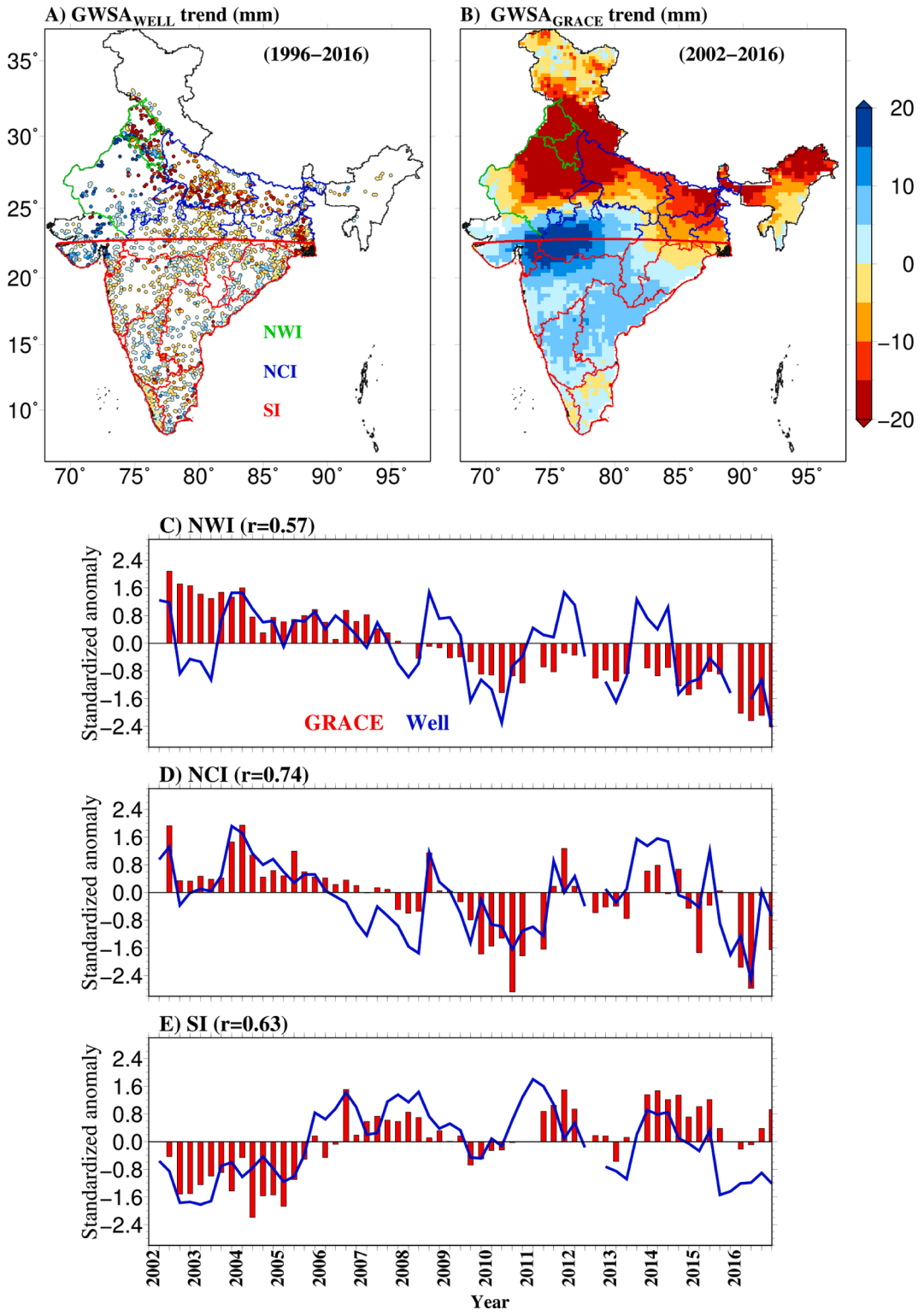


Fig. 3. Groundwater storage anomaly trend from (A) CGWB Well level observations ($GWSA_{WELL}$) and (B) the CSR mascon product from the GRACE satellites ($GWSA_{GRACE}$). (C-E) Comparison of standardized groundwater storage anomaly derived from well level observations and GRACE for north-west India (NWI), north-central India (NCI) and south India (SI). The GRACE RL06 mascon available at 0.25° from CSR (http://www2.csr.utexas.edu/grace/) is used to derive $GWSA_{GRACE}$. The surface water storage anomaly (soil moisture + canopy storage + snow water equivalent) is subtracted from GRACE TWSA to derive $GWSA_{GRACE}$. The surface water storage anomaly is available at 0.25° resolution from Global Land Data Assimilation System (GLDAS) Noah land surface model version 2.

et al., 2017). Hora et al. (2019) pointed out a survivor bias in which missing groundwater stress information from defunct or dry wells in South India might explain groundwater recovery at those locations. Overall, the contrasting changes in the groundwater storage in the north and south India can be attributed to groundwater pumping for irrigation (Asoka et al., 2018; Rodell et al., 2009) and decline in precipitation in the north while an increase in precipitation in South India, respectively.

Droughts cause excess groundwater loss as the extra withdrawal is

Table 1
Chronology of key literature reviews of groundwater depletion in India.

Study area	Time period	Groundwater declining rate	GRACE product, LSM and in-situ data.	Authors (Year)
North-western Indian states (Rajasthan, Punjab, Haryana, Delhi)	August 2002-October 2008	Mean rate of 17.7 ± 4.5 km^3 yr^{-1} (4.0 ± 1.0 cm yr^{-1})	CSR GRACE TWS, GLDAS soil water storage.	Rodell et al. (2009)
Northern India	April 2002-June 2008	54 ± 9 km^3 yr^{-1}	CSR GRACE TWS, 4 hydrological models (Noah, Mosaic, CLM and model from NOAA CPC).	Tiwari et al. (2009)
North-western India	2003–2012	29.4 ± 8.4 km^3/yr (2003– 2007) 20.4 ± 7.1 km^3/yr (2003– 2012)	CSR GRACE RL05 TWS, GLDAS (Noah LSM).	Chen et al. (2014)
Major river basins (Ganga, Godavari, Krishna, Mahanadi)	2003–2012	Ganga: 1.28 ± 0.20 mm/month Mahanadi: No change Godavari: 0.74 ± 0.21 mm/month (↑) Krishna: 0.97 ± 0.21 mm/month (↑)	CSR GRACE RL05 TWS, Soil Moisture, evapotranspiration, runoff from GLDAS (NOAH, VIC and CLM). Also soil moisture from CPC, Precipitation from GPCC, PREC/L), and Evapotranspiration from MODIS.	Soni and Syed (2015)
Northwest India aquifer	2005–2010	3.1 ± 0.1 cm/a (or 14 ± 0.4 km^3/a)	CSR GRACE RL05 TWS, GLDAS (Noah) soil moisture.	Long et al. (2016)
Indo-Gangetic Basin	2000–2012	Net annual depletion of 8.0 ± 3.0 km^3 with declining greater than 1 m yr^{-1} at some locations	Groundwater-level records from 3,429 water wells.	MacDonald et al. (2016)
India	January 2003-May 2014	Haryana, and western Uttar Pradesh: 1.75 to 3 cm yr^{-1} Ganges: 1.25 cm yr^{-1} Punjab: 2.1 cm yr^{-1}	CSR GRACE RL05 TWS, GLDAS (Noah) soil moisture, In-situ groundwater levels from CGWB.	Panda and Wahr (2016)
India	2002–2013	Northern India: 2 cm yr^{-1} & Southern India: 1–2 cm yr^{-1}(↑)	CSR GRACE RL05 TWS, GLDAS (Noah, CLM, VIC, and MOSAIC LSMs), In situ groundwater well observations from the CGWB.	Asoka et al. (2017)
India	2003–2015	Declining trend in northwest India, maximum rate in Delhi: 1.7 cm yr^{-1} Tamil Nadu: 0.7 cm yr^{-1}	CSR GRACE RL05 TWS, CLSM, In-situ observations from CGWB.	Girotto et al. (2017)
India	2003–2014	North India: 1.25 ± 0.14 cm yr^{-1} (−12.56 ± 1.37 km^3 yr^{-1}) & East India: 1.05 ± 0.35 cm yr^{-1} (13.12 ± 4.36 km^3 yr^{-1})	3 GRACE RL05 TWS from CSR, GFZ and NASA JPL, GLDAS (Noah, CLM and VIC LSMs), In-situ observations from CGWB.	Bhanja et al. (2018)

Here all the values shown are depletion rate except those with upward arrows, which signifies increasing groundwater rate in the region.

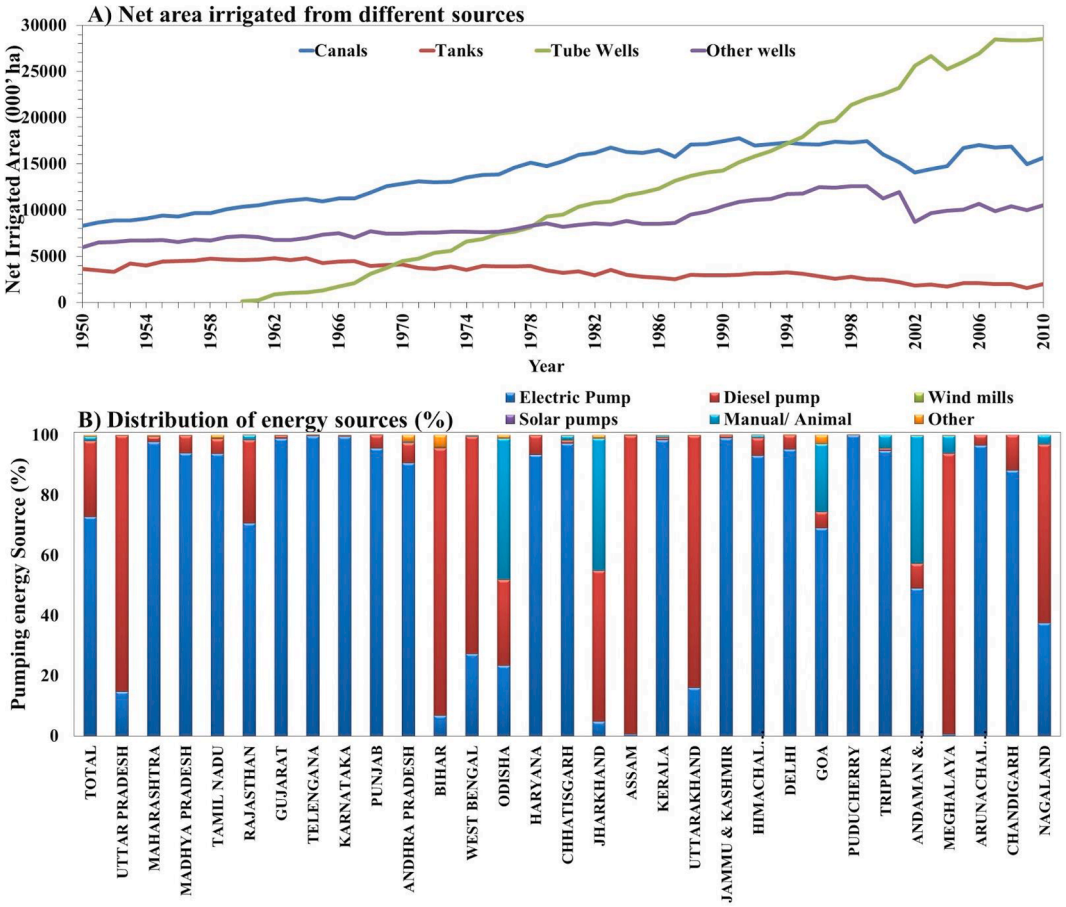


Fig. 4. A) The net area irrigated in India from canal, tanks, tube wells, and other wells during 1950–2010 and (B) the state-wise distribution of different energy sources used for groundwater pumping. The energy sources for groundwater withdrawal is estimated from the census of minor irrigation (http://micensus.gov.in/). Modified from Mishra et al., (2018a)]

needed to meet the drinking water demand and to balance the moisture deficit as seen from the in-situ drawdown groundwater table (Famiglietti and Rodell, 2013; Goldin, 2016; Panda and Wahr, 2016; Taylor et al., 2013). In 2009, precipitation shortage of around 2.5 times the national average had led to the depletion of GRACE TWS and groundwater storage (GWS) of 7.35 cm and 5.63 cm, respectively in northern India (Panda and Wahr, 2016). Depletion rate went up to ~ 80 Gt in 2009 while there is a net rise in groundwater storage in the wet season of 2008 (~24 Gt) and 2011 (~35 Gt) in North-west India (Chen et al., 2014). In 2010, excess rainfall in northwestern parts though below long-term normal was not sufficient to fully recover the groundwater (Panda and Wahr, 2016). In-situ observations show a good agreement (based on correlation) with the GRACE data with the transition from positive to negative anomalies in 2007 with the most remarkable change in 2009 and then recovery in 2011 (Asoka et al., 2017; Panda and Wahr, 2016) [Fig. 3]. An extensive depletion in groundwater storage from the GRACE satellites occurred during the 2002–2016 period that is consistent with the groundwater well observations (Fig. 3). The NWI faced a persistent groundwater drought shown by both GRACE and well observations after 2012 (Fig. 3), which is linked with the pumping of groundwater. North-Central India (NCI) has experienced drought after 2014, mainly due to the decline of long-term precipitation (Mishra et al., 2012; Roxy et al., 2015) as well as pumping.

Extraction of nonrenewable groundwater for irrigation in India has a significant contribution of about 68 $km^3\ yr^{-1}$ for the year 2000, which is the highest in the world (Wada et al., 2012). Mishra et al. (2018) reported that there had been a significant increase in the number of tube wells in India during the 1950–2010 period (Fig. 4). The rise in the number of tube wells considerably outpaced the rise in canals, tanks, and other wells (Fig. 4a). The electric pumps dominate groundwater pumping for irrigation in the majority of states of India (Fig. 4b). Bihar, West Bengal, and Uttar Pradesh have more diesel pumps than electric pumps (Fig. 4b). The northwestern and western states of India rely on abstraction from groundwater wells, canal networks, and subsidized electricity to pump water while the eastern states have fewer canal networks and also unsubsidized pumping facilities (Shah, 2010). Cheap electricity has encouraged farmers to grow water-intensive crops that have led to a rapid decline in groundwater in India (Mishra et al., 2018). The stressed aquifers in India are located in the regions with intensive irrigation, which need urgent technological and policy interventions to ensure water security in the future. While well and GRACE observations help us in identifying the current hotspots of groundwater depletion in India, the country may witness new hotspots in the future due to climate change and increased irrigation and residential water demands.

3. Implications of groundwater depletion

Groundwater depletion is more prominent in north India while satellite-based observations showed an increase in groundwater storage in Peninsular India. Groundwater depletion caused by unsustainable groundwater pumping can have large implications for food and water security in the region. The green revolution led to an increase in food production in India. The irrigated area is on the rise to meet the food demands of the population, which exacerbates the groundwater depletion in the densely populated and highly irrigated parts used from groundwater. More dependence on nonrenewable groundwater to meet food and water supply causes sustainability, environmental, and ecological challenges. In this section, we provide a detailed discussion of the critical implications of groundwater depletion in India.

Food security remains one of India's most profound implications associated with groundwater depletion. Groundwater depletion can be further worsened by the substantial increase in the concurrent hot and dry monsoon extremes are projected in India. Hot and dry extremes during the monsoon season are projected to rise under the warming climate with implications for food production and to maintain crop yields (Aadhar and Mishra, 2020; Mishra et al., 2020, 2021). In addition, area affected by the concurrent dry and wet extremes during the monsoon season have increased and likely to rise in the future. Changes in climate in the monsoon and non-monsoon seasons are projected in India, which can pose tremendous challenges for water availability putting more burden on already depleting groundwater resources. Sowing and harvesting time of crops like rice may vary according to shifts in the monsoon and may affect crop yields (Kelkar et al., 2008). The depletion in root-zone soil moisture caused by the decline in precipitation and air temperature is met by the irrigation mostly from groundwater (Kumar and Mishra, 2020). Dry (October-December) season soil moisture showed a declining trend in the Indo-Gangetic Plain and central India during 1982–2018 (Fig. 5a). However, satellite-based Normalized Difference Vegetation Index (NDVI) from Advanced Very High-Resolution Radiometer (AVHRR) in the same season has significantly increased in large parts of India, including those faced a decline in soil moisture (Fig. 5b). This increase in NDVI is mainly due to irrigation to maintain food production in India, indicating the future needs of groundwater to manage droughts and the impacts of climate variability and change. Notwithstanding the decline in the monsoon season precipitation in Indo-Gangetic Plain and central India, long-term increase in NDVI indicates the use of groundwater based irrigation for food production in India (Ambika and Mishra, 2019; Asoka and Mishra, 2015). The role of groundwater in minimizing the adverse impacts of climate change on food security can increase under the warming climate.

Development of irrigated agriculture led to increased agricultural production and price stability, which are crucial for feeding the world population (Rosegrant et al., 2009). Food demand is also on the rise due to population growth and change in food diets of people enhancing groundwater depletion (Foley et al., 2011). India is feeding its population by over-drafting of groundwater from aquifers and may lose its capacity of food production in the coming decades. India faces blue water scarcity during the crop growing period, and irrigation water demands are primarily met by the unsustainable abstraction of groundwater (Rosa et al., 2020). Also, about 11% of nonrenewable groundwater use for irrigation is associated with international food-trade, and India is among the top three contributors (Dalin et al., 2017). Water scarcity has implications on the expansion of irrigated areas as well as slowing down the yield growth rate of cereals in developing countries (Rosegrant et al., 2009). Therefore, proper usage of agricultural water forms the basis of solution for water crisis assessments (Rosegrant et al., 2009; Zhu et al., 2019).

While groundwater depletion poses challenges for food and water security in India, it impacts the environment and ecosystems in different ways. The environmental implications of groundwater depletion are well beyond decreasing freshwater availability as it results in degradation of groundwater quality (Famiglietti, 2014) and an increase in the pumping cost from deeper depths (Konikow and Kendy, 2005). Moreover, groundwater depletion contributes to sea-level rise, a decrease of groundwater discharge to streams, land subsidence (Giordano, 2009), and seawater intrusion in coastal aquifers (Aeschbach-Hertig and Gleeson, 2012; Konikow and Kendy, 2005). Loss of groundwater results in decline in surface water (Condon and Maxwell, 2019) as rivers like the Indus and Colorado now do not reach the oceans (Famiglietti, 2014). Baseflow to Ganga river is reduced in summer due to groundwater pumping (Mukherjee et al., 2018). Shallow groundwater depletion due to pumping and land-use change are the likely drivers for reduced streamflow in the human-dominated watershed of Arkavathy sub-basin in southern India during the last fifty years (Penny et al., 2020; Srinivasan et al., 2015). Groundwater over-extraction in the agricultural regions outside the forest may result in deeper groundwater levels in an adjacent basin, which can influence streamflow (Krishnaswamy et al., 2013; Sekhar et al., 2004). Streamflow declines not only near the areas of groundwater pumping but also have a persistent effect in the downstream regions (Condon and Maxwell, 2019). In addition, groundwater vegetation interactions are important for deep rooted forest ecosystems as they have ability to groundwater uptake significantly from higher

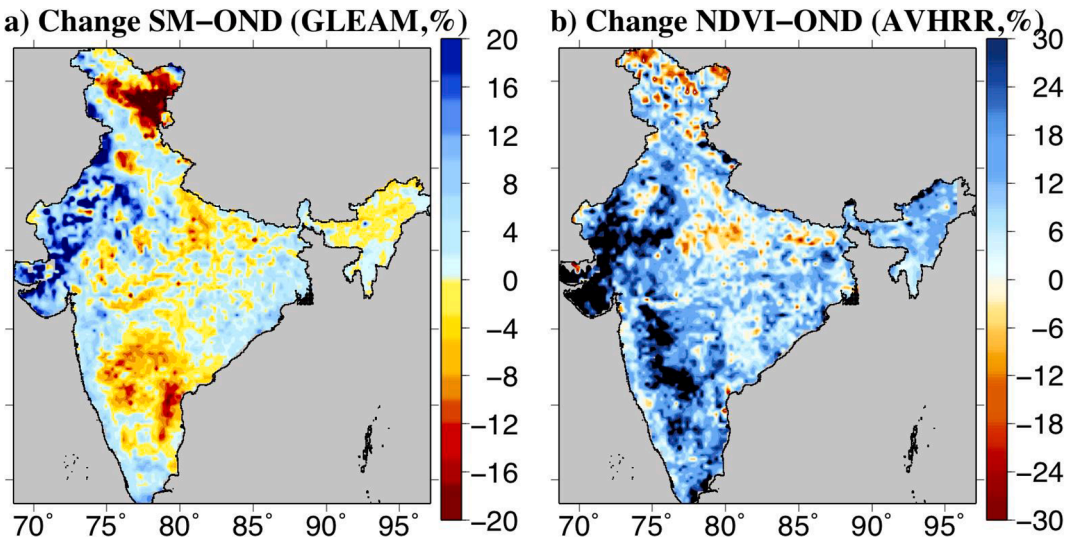


Fig. 5. (a) Changes in soil moisture from Global Land Evaporation Amsterdam Model (GLEAM) during October-December (SM-OND), (b) changes in Normalized Difference Vegetation Index (NDVI) from Advanced Very High-Resolution Radiometer (AVHRR) during October-December for 1981–2018 period. Changes were estimated using Mann-Kendall trend and Sen's slope method.

depth to distribute to short rooted vegetation and can play key roles in ecosystem and also have implications for groundwater recharge (Orellana et al., 2012).

Groundwater depletion can have ecological implications. For instance, riparian and aquatic ecosystems also get affected by groundwater storage variability in days to centuries timescales (Barlow and Leake, 2012), especially during the low flow season (Gleeson & Richter, 2018). de Graaf et al. (2019) showed that the hotspots areas of groundwater depletion of Upper Ganges and Indus basins have already reached their limits to environmental flow before 2010. The southeastern parts of the Ganges basin have not reached their environmental flow limit though experienced significant head declines (de Graaf et al., 2019). The reason may be the delay between the effects of pumping on streamflow and disjoint between the riverbed and groundwater level (de Graaf et al., 2019). Dry season low flows could have negative impacts on Ganges river endangered dolphin species (Choudhary et al., 2012; Dey et al., 2019). The Krishna river basin is an example of both ecological destruction and drought-driven conflict (Srinivasan et al., 2012). Groundwater pumping can crucially influence ecosystem services while sustainable groundwater resources management can strengthen ecosystem resilience for future climate extremes and variability (Qiu et al., 2019). Ferguson and Gleeson (2012) reported that coastal aquifers are relatively more vulnerable to the abstraction of groundwater than the sea level rise projected under different hydrogeologic conditions. Overall, groundwater depletion in different parts of India can have various short and long-term environmental consequences. However, the role of groundwater depletion on environment and ecology has not been widely studied in India.

4. Sustainable management of groundwater resource

Considering the rapid depletion of groundwater for irrigation and its implications for food and water security in India, sustainable management of groundwater resources is needed. Sustainable management of groundwater resources in India needs an integrated approach that reduces groundwater abstraction and increasing the groundwater recharge. Since groundwater pumping happens primarily for irrigation, changes in the cropping patterns, improving the irrigation water use efficiency, shifting to the crops that are less water-intensive, and other policy interventions can be potential areas that need attention. On the other hand, enhancing the groundwater recharge during the monsoon season is required, which needs information related to aquifer characteristics, rainwater harvesting, managed aquifer recharge, and water quality. In this section, we discuss the ways that can lead to the sustainable management of groundwater resources in India.

Irrigated agriculture is the largest consumer of groundwater in India. Sustainable and efficient agriculture methods are needed to cope with the pressure of increasing crop production with less water availability (Lenton, 2014). Foley et al. (2011) stated that the world's agricultural yield could be derived within 95% for 16 major crops by calculating the distribution of potential new calories globally. India is one of the hotspots of yield stagnation for maize, rice, wheat and soybean cereals, which is a challenge to meet the agricultural demand (Ray et al., 2012). Shifting of growing areas of crops like rice and wheat can increase the net farmer's revenue as well as reduce the groundwater and energy consumptions in already stressed regions of India (Devineni and Perveen, 2012). Davis et al. (2018) found that consumptive irrigation water demand is mainly due to the production of wheat during the Rabi season. Rice is not a water-efficient cereal; therefore, diet shift from rice to other alternatives like maize or millet can reduce the irrigation water demand by 33% (Davis et al., 2018). Switching to lower water demand crops like from rice to pulses may help in reducing the groundwater depletion rate in India, as shown for Punjab (Russo et al., 2015).

Improving the irrigation water use efficiency can play a crucial role in groundwater sustainability in India. Direct seeding of different plant varieties on level land with furrow diking by energy-efficient sprinklers and knowing when and where to irrigate can increase the agricultural water use efficiency (Gleick, 2003). The adoption of low-cost tensiometer based irrigation scheduling can save groundwater used for irrigation and power usage leading to economic advantage in Punjab, India (Vatta et al., 2018). Already existing strategies that can help in groundwater conservation include drip irrigation, remedies in soil, restoration of wetland, and water-reducing crops (Foley et al., 2011). Allocation of irrigated water between crops can increase the economic efficiency of water usage as well as alleviate irrigation water consumption (Bierkens et al., 2019). Crop growing patterns, methods, and efficiency in using water decide the rate of groundwater abstraction (Siebert et al., 2010). In the western Gangetic basin, Fishman et al. (2015) showed that maximum theoretical agricultural efficiency can limit the depletion but cannot reverse the depleted state. Therefore, aquifer stress may not be reduced by improving efficiency in the regions of combined surface and groundwater usage due to reduction in irrigation return flow (Wada & Heinrich, 2013). Moreover, farmers can irrigate more land by the water saved from irrigation efficiency

(Fishman et al., 2015; Grafton et al., 2018; Perry et al., 2017). Flood irrigation leads to more evapotranspiration losses, which increases aquifer depletion and results in soil salinization (Giordano, 2009; Scanlon et al., 2012). Optimized use of combined surface and groundwater for irrigation (Konikow and Kendy, 2005) can be productive, which provides pathways to minimize aquifer stress and sustainable groundwater development (Wada & Heinrich, 2013). However, this can be limited to the regions that have considerable surface water availability during crop growing seasons.

Enhancing the groundwater recharge is vital for the long-term sustainability of groundwater storage in India. Shallow and unconfined aquifers are recharged during the monsoon seasons (southwest and northeast monsoons) in India. Managed aquifer recharge (MAR) can be seen as an adaptation measure to meet the increasing future water demands and to restore the groundwater level (Dillon et al., 2019; Massuel et al., 2014; Shah, 2014; Stefan and Ansems, 2018). Modifications of streambed channels, bank filtration, water spreading, well recharge, and release from the reservoir are a few examples of managed aquifer storage (Dillon et al., 2019). Watershed development (WSD) in India is aimed to increase rainwater storage and agricultural yields to meet food security objectives (Gray and Srinidhi, 2013). WSD can help in reducing the impact of declining groundwater yield by supplemental recharge of groundwater in drought-prone hard rock aquifers in Karnataka (Chandrakanth et al., 2004). However, WSD is constrained by limitations in regular data collection and impact assessments regarding drought resilience (Gray and Srinidhi, 2013). Aquifer-based common pool resource (Sophocleous, 2010) is becoming popular for managing groundwater in India (Kulkarni et al., 2015; Saha et al., 2018). Water can be stored in subsurface reservoirs instead of surface reservoirs to restrict the evaporation losses (Aeschbach-Hertig and Gleeson, 2012). Canal networks can be converted to artificial recharge systems in India (Shah, 2009). Canal links help maintain groundwater levels through recharge while also providing irrigation water when required to reduce groundwater abstraction and alleviate irrigation costs. Also, during the monsoon season, on-farm water storage and rainwater harvesting can enhance the groundwater recharge and reduce the pressure of groundwater system during the dry season (Aeschbach-Hertig and Gleeson, 2012; Glendenning et al., 2012; Rockström et al., 2010). Water harvesting system accounts for 50% of aquifer recharge in the semi-arid watershed of southern India (Perrin et al., 2012). Groundwater banking (Maliva, 2020) may offer sustainable groundwater management by providing storage during extreme floods and then abstracting that water during long term droughts (Scanlon et al., 2012). Conjunctive use of canal and groundwater is common in scarce canal water and marginal saline groundwater condition in the arid and semi-arid regions of northwest India (Mandare et al., 2008). Coordination between surface water and groundwater monitoring agencies and framework is needed for long-term quantifications and allocation of water resources (Srinivasan and Kulkarni, 2014).

Water governance such as subsidizing electricity will increase groundwater pumping, as seen in India (Suhag, 2016) and even in the areas where surface water is available and during the monsoon season. Rapid groundwater withdrawal in northwest India is not sustainable for the long-term management of groundwater resources (Lenton, 2014). A significant abstraction of groundwater for irrigation leads to lowering of the water table, which may require deepening of tube wells. However, the cost associated with deepening the tube wells may not be a problem for wealthy farmers with electricity subsidies. For instance, the economic value is not a significant concern for medium and large farmers in Punjab (Jain, 2006; Sarkar, 2012) where wells are dug deep to reach over 40 m for groundwater, so most probably the environmental limit will decide the groundwater availability (de Graaf et al., 2019). Tariff policies based on groundwater exploitation will help in the sustainable use of groundwater. In northwestern and western India, metered tariff policies to promote groundwater conservation while in eastern parts like West Bengal, hybrid flat-cum-metered tariff should be done to benefit the farmers as well as to check the unsustainable exploitation of groundwater (Sidhu et al., 2020). However, policy implementation to curb groundwater exploitation is controlled by large holding farmers, communities, and governments (Shiferaw et al., 2008), as shown by the contrasting cases of success in West Bengal and failure in Gujarat states (Mukherji, 2006). In the hard rock areas, some initiatives are started by farmers, communities, and NGOs to supplement groundwater recharge and counter depletion rate by enforcement of rules (Shah, 2012).

Policy and state-level incentives will drive the crop use pattern and related irrigation water usage in India (Davis et al., 2018; Shiferaw et al., 2008). Minimum support prices for some cereals like rice and wheat by the government has caused them to grow even in unlikely areas (Chatterjee and Kapur, 2017). Fishman et al., (2011) showed that deep alluvial aquifers in northwestern India (Punjab), agriculture is energy and land limited as water is extracted from deeper depths. On the other hand, in central India (Telangana), the agriculture is water-limited as the shallow hard rock aquifers have small storage capacity that is depleted and renewed by rainfall. Therefore, management of groundwater in alluvial aquifers is a long-term sustainability problem based on energy usage while in hard rock aquifers it is a short-term reliability problem (Fishman et al., 2011). Regional strategies selection is critical for policy implementation and adaptation to strengthen regulation and management of groundwater and irrigation considering local scale heterogeneity in alluvial and hard rock aquifers (Edwards and Guilfoos, 2020; Giordano, 2009; Sophocleous, 2010; Theesfeld, 2010). Solar irrigation pumps (SIP) can be the alternative policy to help mitigate groundwater depletion in water-scarce regions (Shah et al., 2018). However, using solar pumps as a standby option to diesel or electric pumps result in more groundwater pumping (Durga et al., 2016; Gupta, 2019; Shah et al., 2018), and can be promoted if farmers pay the price for each unit of extracted groundwater (Agrawal and Jain, 2019). O'Keeffe et al. (2018) found that changes in farmers' irrigation behaviour could impact water consumption. The interactions and feedback between social practices and environment should be understood to make a better policy for water security perspective (Giuliani et al., 2016; Gleeson et al., 2012a; Li et al., 2017; Sivapalan et al., 2014; Wagener et al., 2010).

5. Critical needs for understanding the groundwater storage variations

Measurement and modelling are crucial for better understanding the groundwater depletion, implications, and sustainable management. Given that India is the frontrunner in the massive groundwater extraction, we need to identify the areas that need enhancement for the better understanding of the groundwater storage variability and groundwater recharge. In-situ and satellite-based monitoring play a vital role in examining the role of natural climate variability and anthropogenic influence on groundwater storage in India. On the other hand, enhancing the groundwater recharge requires the methods of groundwater recharge estimations and understanding of the aquifer characteristics. Understanding the observed and projected impacts of climate change are vital for the sustainable management of groundwater resources in India. In this section, we review a few critical needs that can help us in better understanding groundwater depletion and finding out the ways for sustainable management.

5.1. Groundwater measurements and monitoring

5.1.1. In-situ well observations

Measurement and monitoring of groundwater storage are essential for understanding the anthropogenic and natural impacts. Central Ground Water Board (CGWB) has developed a network of 23,125 observation wells all over the country as of March 2017 (CGWB, 2019) (Fig. 6). Groundwater well measurements are available four times in a year in January, May, August, and November. The pre-monsoon well

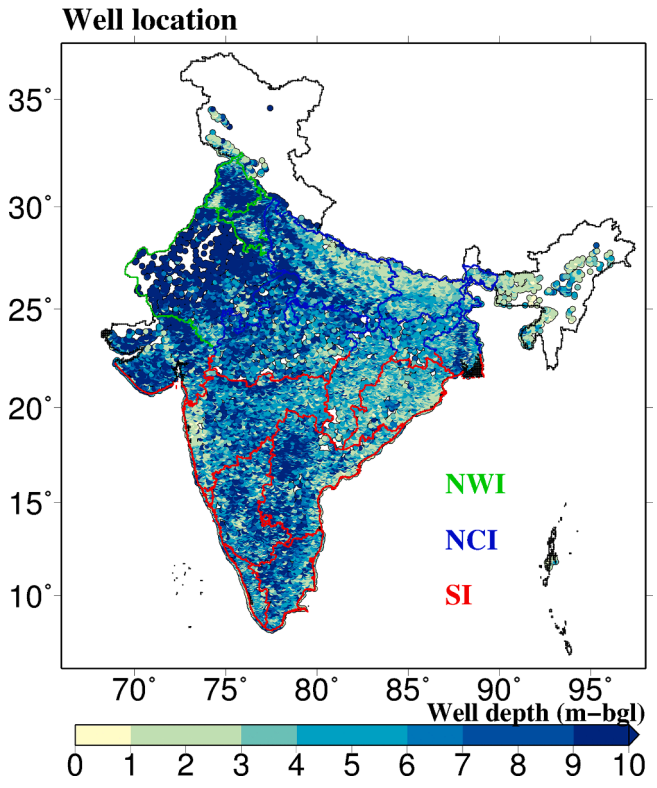


Fig. 6. Location of groundwater observational wells from the Central Groundwater Board (CGWB) and depth to water table (m) below ground level (m-bgl). Color polygons show northwest India (NWI, green), northcentral India (NCI, blue), and south India (SI, red). Color scale shows water table depth below ground level (m-bgl). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

level data is monitored in March, April, or May depending on the monsoon onset in a given region. For most of the states in India, pre-monsoon well observations are in May, for West Bengal, Orissa, and Kerala, observations are taken in April and for the north-eastern part in March as the summer monsoon onset occurs earlier (CGWB, 2014). Groundwater well observations for the post-monsoon are taken in November to assess the impact of monsoon on groundwater. The water level in November reflects both the groundwater recharge and withdrawal components, while the water level in January reflects the withdrawal for irrigation during the Rabi (November-March) season. Groundwater well level in August shows the influence of the monsoon season groundwater recharge in the well observations (Asoka et al., 2018).

While CGWB monitoring wells are located across India, there are issues and challenges associated with in-situ measurements of groundwater. We do not know if the current monitoring of the groundwater level is sufficient to understand space–time variability in groundwater storage in India. Therefore, the current density of groundwater wells in different regions has to be evaluated. Designing of a groundwater monitoring network involves two components (Zhou, 1996): number of observation wells and their locations (density of the network), and the number of observations per unit time (sampling frequency). There have been studies focusing on the optimization of groundwater monitoring networks (Ohmer et al., 2019). The majority of approaches are based on the geostatistical analysis (Dhar, 2013; Kumar et al., 2005; Prakash and Singh, 2000; Yang et al., 2008) such as kriging to interpolate the groundwater-surface to identify the additional requirement of observation wells or redundancy in the existing wells. Algorithms based on Kalman filter (Wu, 2004; Zhang et al., 2005) have been used for the assessment of groundwater network density and measurement frequency, entropy (information) theory for optimizing information and cost (Mogheir and Singh, 2002; Mondal and Singh, 2012) in the groundwater monitoring network. CGWB monitors groundwater level four times in a year, and it remains unclear if the temporal resolution of existing CGWB observations is sufficient for examining the drivers of groundwater depletion in India. The groundwater measurement sampling frequency of once every month is desired for detectability of trends, accuracy in the estimation of groundwater fluctuations, and accuracy of measurement of stationary components (Zhou, 1996).

Adequacy of groundwater monitoring wells can vary from one region to the other depending on the spatial variability and heterogeneity in groundwater (Fig. 6). Monitoring wells can be additionally placed in the areas of high variability; for instance, in intensively irrigated areas increased density of observational wells to get a more robust estimation, which can better capture the regional variability. In contrast, there might be regions in the country that may have redundancy in the groundwater monitoring network, which also need to be identified. An average density of 1 well in every 0.25 km^2 area, depending on the hydrogeology of hard rock aquifers is necessary (Kulkarni and Shankar, 2009). Locating the ideal place for monitoring well is an essential choice for groundwater managers, mostly because of the cost related to groundwater monitoring. Similarly, we need to examine the desired frequency of groundwater observations based on groundwater fluctuation and can be improved to better understand the role of irrigation during the crop growing seasons on groundwater storage variability. Community-based monitoring presents a way to complement existing groundwater monitoring, where community volunteers measure the regional water levels and upload through a web-based portal that can be used by water resource planners (Little et al., 2016). Citizen science approach can provide a valuable data set regarding the monthly water level measurements and can cover regions of high spatial variability. Overall, there is a need for careful evaluation of the appropriate spatial and temporal resolutions of groundwater well observations in India to examine the role of natural and anthropogenic factors on groundwater storage.

5.1.2. *Satellite-based measurement*

Other than in-situ observations of groundwater storage variability, the satellite-based information has played an essential role. GRACE satellite gravimetry is the first remote sensing-based technique that can be used to estimate large scale changes in groundwater storage (Rodell et al., 2007; Yeh et al., 2006). Groundwater depletion due to anthropogenic and climate-driven fingerprints can be estimated using the groundwater storage anomalies calculated using the GRACE satellites (Pokhrel et al., 2012, 2015; Richey et al., 2015b; Scanlon et al., 2012; Wada et al., 2010). The GRACE satellites fill the gaps that exist in the well level observations, however, at much coarser spatial resolution. For instance, GRACE satellites capture the groundwater storage variability of shallow and deep wells (Meghwal et al., 2019; Tiwari et al., 2009). The GRACE satellite measures the vertically integrated TWS anomaly monthly with an accuracy of 1.5 cm equivalent water height (Famiglietti and Rodell, 2013) and spatial resolution of roughly 400 km (Tapley, 2004) depending on the altitude of GRACE satellites and the distance between the twin satellites.

Apart from the coarse spatial resolution, the GRACE based groundwater estimates can have other sources of uncertainty. There are different products available from the GRACE satellites that contribute to considerable uncertainty in the estimation of TWS, depending on the region (Fig. 7). GRACE-TWSA from JPL and CSR mascons captures groundwater changes better than spherical harmonics (SH) primarily in the regions affected by groundwater depletion in India (Save et al., 2016). Spherical harmonics suffer from more signal loss in the pre-processing (Scanlon et al., 2016) while leakage errors in GRACE mascons are relatively lower. The GRACE mission has been successful in assessing the groundwater depletion rates, especially in the northern parts of India. However, there is substantial uncertainty which resulted

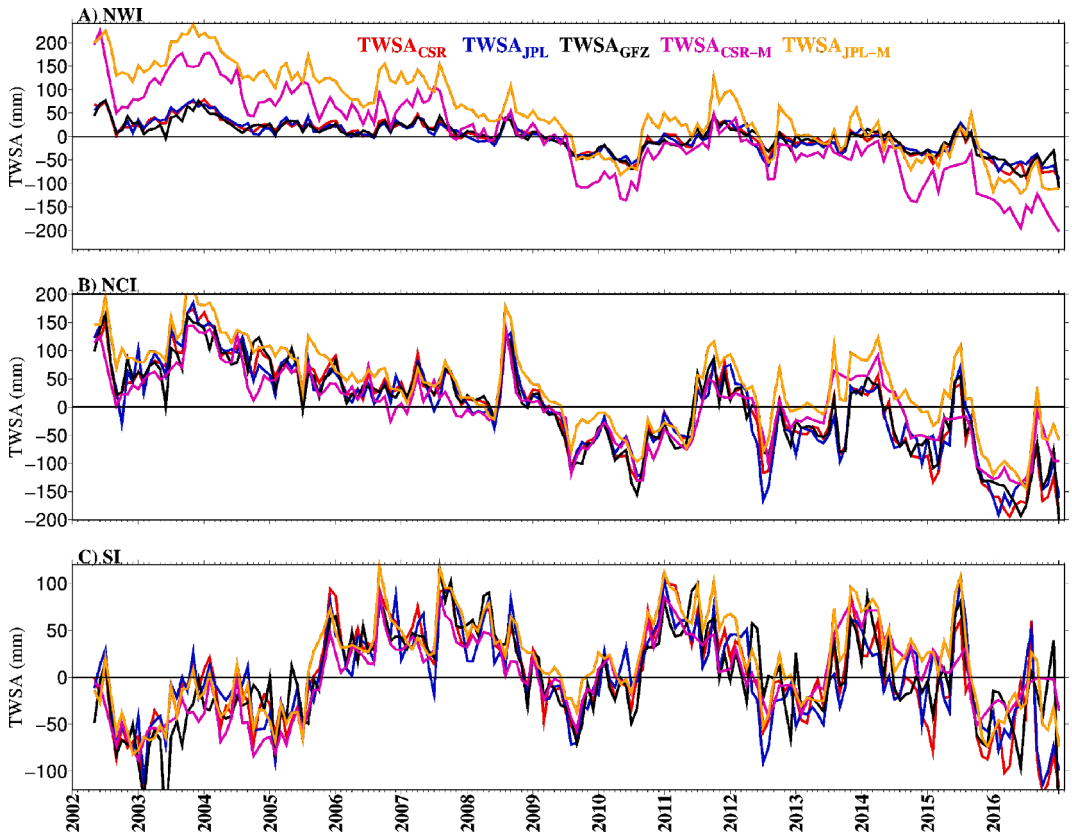


Fig. 7. Comparison of Terrestrial Water Storage Anomaly (TWSA) from different GRACE solutions for the A) northwest (NWI), B) northcentral (NCI), and C) South India (SI). The monthly mean is removed to de-seasonalized TWSA time series. Three RL05 spherical harmonics solutions and two RL06 mascon solutions are used. The spherical harmonic solutions from Centre for Space Research (CSR) at the University of Texas, NASA's Jet Propulsion Laboratory (JPL), and Deutsches Geo-Forschungs Zentrum (GFZ) were used after applying a scaling factor (TWSA$_{CSR}$, TWSA$_{JPL}$ and TWSA$_{GFZ}$). The two mascon solutions are from JPL (TWSA$_{JPL-M}$) and CSR (TWSA$_{CSR-M}$).

in an overestimation of the groundwater depletion in northwest India (Long et al., 2016). Scale factors estimated from comparing land surface model simulated TWS changes (Landerer and Swenson, 2012) are used for leakage bias correction in GRACE estimates (Velicogna and Wahr, 2006). Scaling factor method is suitable for restoring seasonal TWS trends and not for the interannual variation due to uncertainty in land surface models (Chen et al., 2016). Moreover, as most of the CGWB observations are from shallow wells, care must be taken to compare the groundwater storage variability from the GRACE and in-situ observations (Meghwal et al., 2019). GRACE based groundwater storage estimates are more accurate for the large areas as limited spatial resolution and noise could cause bias and leakage errors if the region is small (Longuevergne et al., 2010; Scanlon et al., 2016). This also highlights the need for in-situ observations in small irrigated basins, which cannot be accurately measured using GRACE satellites.

GRACE groundwater storage anomalies have limitations associated with coarser resolution, which can be overcome by a statistical downscaling approach using evapotranspiration (Yin et al., 2018), however only in the regions that have a strong correlation with groundwater storage anomalies. Moreover, if the local scale variability is not consistent with the regional pattern, then the downscaling will not capture the changes in groundwater level. Downscaling of the GRACE data can be further validated using the in-situ measurements at the same spatial scale (Sun, 2013). Then GRACE monthly measurements can help in filling the temporal gaps in quarterly groundwater levels measurements in India. Downscaled TWS using artificial neural network (ANN) (Miro and Famiglietti, 2018) can capture the variability present in the combined effects of climatic and anthropogenic impacts (Seyoum and Milewski, 2017). However, the limitation of such data-driven models is that they are mostly domain-specific and have to be recalibrated to apply for a different region of interest. These kinds of studies will be useful in the areas where groundwater level measurements are sparse. GRACE Follow-On (GRACE-FO) mission launched in 2018, will continue measuring the movement of masses to provide valuable estimates (Ciracì et al., 2020; Flechtner et al., 2016; Kusche et al., 2016) with laser ranging system to improve the spatial resolution (McCabe et al., 2017).

5.2. *Land surface modelling (LSM) and statistical approaches*

Global hydrological and land surface models played a vital role in improving our understanding of climatic and anthropogenic linkage with terrestrial and groundwater storage variability. Several land surface models have taken human factors like groundwater withdrawal into account while considering the water budget (Pokhrel et al., 2016; Sood and Smakhtin, 2015). For instance, WaterGAP (Döll et al., 2012), PCRaster Global Water Balance (PCR-GLOBWB) [(Wada et al., 2010)], H08 (Hanasaki et al., 2018), Community Water Model (CWatM) (Burek et al., 2019), LPJmL (Rost et al., 2008), WBMplus (Wisser et al., 2010), MATSIRO (Pokhrel et al., 2012), and HiGW-MAT (Pokhrel et al., 2015) consider human influence in simulating water budget and land surface processes. These models estimate long-term groundwater recharge and simulate groundwater abstraction influence on groundwater storage variability at regional and local scales, which cannot be quantified by the GRACE satellites due to coarser spatial resolution (Döll et al., 2014). PCR-GLOBWB has been coupled with MODFLOW groundwater model at a high resolution (Wada et al., 2016) to simulate groundwater levels and lateral groundwater flow and they found a substantial impact of humans on terrestrial water fluxes in India. Groundwater storage estimates,

therefore, can be estimated using the GRACE TWS and surface water storage estimated from the land surface models.

The accuracy of land surface model estimates depends on the input meteorological forcing data and model parameterization of the physical processes. Since the GRACE cannot separate various components of TWS change, water storage changes in snow, surface water reservoirs, and soil have to be estimated from the other sources (e.g. land surface models). The uncertainty associated with water storage components simulated by the land surface models can influence groundwater storage anomaly estimates from the GRACE satellites (Chen et al., 2016; Long et al., 2016; Richey et al., 2015a; Rodell and Famiglietti, 2002; Van Loon et al., 2017). Scanlon et al. (2018) compared decadal trends in land water storage from seven global water models and reported that models underestimate water storage trends against those estimated from the GRACE satellites mainly due to lack of human intervention representation and climate forcing uncertainty in most of the models. However, the estimates of groundwater depletion using hydrological models are generally higher in the irrigated regions of Northern India (Konikow, 2011; Wada et al., 2012) when compared with GRACE estimates of Rodell et al. (2009) and Tiwari et al., (2009), as the models mostly do not consider the return flow from irrigation, diversions and lateral groundwater flow (Aeschbach-Hertig and Gleeson, 2012). A better understanding of the physical processes in the hydrological models can provide improved estimates of the groundwater changes in the current and future climate.

A proper representation of subsurface characteristics in the models is required to quantify better the groundwater response times, recharge rates, and groundwater-climate feedbacks. Groundwater-atmosphere feedback is linked to human water use, which is influenced by groundwater pumping for irrigation (Keune et al., 2018). Lateral groundwater flow can aggravate the water table pattern but can buffer the seasonal variability of groundwater levels (Zeng et al., 2018). Therefore, the inclusion of lateral groundwater flow in the earth system models (Condon and Maxwell, 2019) can be helpful in simulating terrestrial water fluxes (Maxwell and Kollet, 2008). Parameterization of groundwater capillary flux in global hydrological models has a strong dependence on water table depth and hydraulic properties of soil, and changes recharge and evapotranspiration (Koirala et al., 2019). Hydraulic network connectivity decreases with depth in the fractured layer (Guihéneuf et al., 2014). Subsurface properties can influence river discharge as well. For instance, the Himalayan rivers have transient groundwater storage of around 45 days (Andermann et al., 2012). Therefore, the inclusion of subsurface heterogeneity in the modelling studies can further enhance groundwater recharge rates and sensitivity of recharge to climate variability (Hartmann et al., 2017). Better sub-surface representation in land surface models can improve groundwater persistence, which can be taken into account for management strategies to climate change impacts (Cuthbert et al., 2019).

Data assimilation of GRACE observations along with the inclusion of anthropogenic modules like groundwater abstraction in hydrological models can improve simulations of groundwater storage variability (Girotto et al., 2017; Li et al., 2019; Nie et al., 2019). Also, GRACE data assimilation can be useful to make robust decisions in monitoring groundwater-related droughts (Famiglietti, 2014; Famiglietti and Rodell, 2013; Mishra and Singh, 2010; Thomas et al., 2017). Girotto et al., (2017) assimilated GRACE data in Catchment Land Surface Model (CLSM) for India and reported an improvement in the simulation of groundwater against the in-situ observations. In the irrigated areas of northwestern India, inter-annual variability in groundwater trends are more reasonable in assimilation estimates but also shows an erroneous negative ET trend due to lack of anthropogenic representation like irrigation and groundwater extraction in the CLSM model (Girotto et al., 2017). Joint assimilation of GRACE and Soil Moisture and Ocean Salinity (SMOS) data can improve the TWS estimates especially soil moisture and groundwater components (Tian et al., 2017) and, therefore, can capture enhanced ET due to increased soil moisture from irrigation. Also, assimilation of the GRACE data and model simulations incorporating groundwater extraction for irrigation can provide better estimates of TWS and evapotranspiration (Nie et al., 2019).

Other than the land surface modelling-based approaches to estimate groundwater storage variability, statistical and data-driven approaches have also been used. Among the data-driven approaches, Bayesian networks (Fienen et al., 2013), response matrices (Cheng et al., 2011), polynomial chaos (Laloy et al., 2013), and hierarchical based method (He et al., 2013) have been used for surrogate groundwater modelling (Asher et al., 2015). Time series modelling of groundwater with response functions can give insights into the variation of groundwater levels and recovery time from drought and quantification of aquifer stressors (Bakker and Schaars, 2019). Nonlinear transfer function noise (TFN) model can be used to model groundwater hydrographs for simulating episodic recharge and head declines (Chappell et al., 2017; Peterson and Western, 2014) and to statistically interpolate monthly groundwater hydrographs to get regular frequency at daily time steps with temporal kriging (Peterson and Western, 2018). van Dijk et al. (2020) used a TFN model in northwestern India and found that geological variability in the aquifer system affects the aquifer response to its drivers, which can be used in mitigation strategy. Random forests (machine learning technique) can use physically-based model predictions to predict shallow groundwater table trends while using geological, topographical, land cover, precipitation variables as covariates (Koch et al., 2019). A comprehensive review of artificial intelligence (AI) methods for groundwater level modelling is given in Rajaee et al. (2019). Support vector regression (SVR) performs better than the ANN when TWSA is used as a significant variable to predict groundwater level in India (Mukherjee and Ramachandran, 2018). Similarly, satellite-based vegetation cover as an indicator of groundwater storage (Bhanja et al., 2019a; Krishnaswamy et al., 2014) is used in natural vegetation areas in India where the groundwater table is shallow and scarce in-situ observations. Support vector machine (SVM) generated irrigated classified cropland maps for crop seasons show the spatiotemporal evolution of irrigation in the Kabini river basin in south India to identify the multi-cropping and intensive agricultural practices and their impacts on groundwater resources (Sharma et al., 2018).

Considering the rapid depletion of groundwater in India, efforts to integrate modelling, assimilation of GRACE data and statistical approaches are required. Moreover, a country-level high-resolution groundwater model should be developed to simulate the groundwater responses to drought (Hellwig et al., 2020). Model improvements can be made by including high-resolution parameterization and coupling of gradient-based lateral groundwater flow into the hydrological models (Hellwig et al., 2020). Water demand and usage data should also be incorporated into simulations of groundwater variability (Massoud et al., 2018). Old boreholes and pumping wells act as recharge point sources. Therefore, their inclusion in the hydrogeological models will enhance groundwater levels estimates (Ballukraya and Srinivasan, 2019). Agricultural and irrigation systems in India are complex, and representing them in the modelling framework remains a challenge. For instance, most of the global hydrological models do not represent double or triple cropping systems (Fig. 1a) that exist in India. Moreover, the heterogeneity in Indian agriculture due to small landholdings cannot be properly represented in the coarse resolution hydrological models. The amount of groundwater pumping in the models may vary based on the irrigation schemes and choice of crops. Therefore, the models need to be refined to estimate the influence of agriculture on groundwater storage variability in India.

5.3. Estimation of groundwater recharge

Estimation of the groundwater recharge is critical for the sustainability of the groundwater resources. Groundwater recharge can be estimated either by water table fluctuation (WTF) or water balance (WB) methods (Asoka et al., 2017; Bhanja et al., 2018; Healy and Cook, 2002;

Kinzelbach et al., 2002; Scanlon et al., 2002; Sophocleous, 1991). The WTF method is based on the principle of change in the water table in an unconfined aquifer (Healy and Cook, 2002). The water level fluctuations are studied and mapped for each monitoring period indicating the frequency distribution of seasonal, annual, decadal variations. Long-term observations are used (Asoka et al., 2017) for groundwater recharge estimation, which is free from missing values and other inconsistencies (Hora et al., 2019). Specific yield provided by the CGWB, or high-resolution gridded map (0.1° x 0.1°) (Bhanja et al., 2016), or Indo-Gangetic Basin (IGB) aquifer mapping (MacDonald et al., 2016) can be used to estimate groundwater recharge in India. Groundwater recharge can be estimated by multiplying specific yield and the difference in the groundwater level (Asoka et al., 2017; Bhanja et al., 2016). The water balance method can also be used for groundwater recharge estimation, which follows the conservation of mass of water. Therefore, groundwater recharge rate (R) can be estimated from the water budget equation $R = P - ET - \Delta SM - SR$, where P, ET, ΔSM and SR are precipitation, evapotranspiration, soil moisture change, and surface runoff, respectively (Bhanja et al., 2018; Healy and Scanlon, 2010). Double water table fluctuation (DWTF) method is used in the hard rock and semi-arid regions to estimate specific yield and groundwater recharge (Maréchal et al., 2006). In the DWTF method, first, the specific yield is estimated from groundwater budget and water level changes during the dry season, and then recharge during the wet season can be calculated from WTF method (Maréchal et al., 2018). The combined use of electrical resistivity tomography and magnetic resonance sounding is efficient for estimating groundwater recharge in hard rock aquifers in Western Ghats of Southern India (Descloitres et al., 2008). The multi-tracers study used in the Indo-Gangetic alluvial aquifers reveals that local flow conditions are disturbed due to intense abstraction in deep groundwater and may significantly improve meteoric recharge to depth (Lapworth et al., 2015). Also, global hydrological models [CLM 4.0 (Richey et al., 2015b), WaterGAP (Döll and Fiedler, 2008), PCR-GLOBWB (Wada et al., 2010), H08 (Hanasaki et al., 2018)] can be used to estimate the groundwater recharge. For instance, Asoka et al., 2017; Bhanja et al., 2018 estimated groundwater recharge from groundwater wells using the WTF method and also using the PCR-GLOBWB hydrological model.

Direct measurement of groundwater recharge is difficult, and its spatial and temporal variations increase the complexity (Healy and Scanlon, 2010). Both water table fluctuation and water balance methods have limitations. For instance, Healy and Scanlon (2010) reported that recharge estimation from the WTF method is the best for shallow aquifers and cannot be applied to confined aquifers where fluctuation can be minimal. A majority of the CGWB wells in India are placed in shallow aquifers (Asoka et al., 2017; Hora et al., 2019). Therefore, recharge estimates may not provide information on the influence of irrigation. Moreover, there can be uncertainty regarding the choice of specific yield as uniform values can produce bias in groundwater recharge estimates in the WTF method (Asoka et al., 2018; Healy and Cook, 2002). The main limitation of the WB method is the lack of measured hydrogeologic parameters like porosity and permeability in the estimation as it is based on precipitation rates (Bhanja et al., 2018). Inter-aquifer flow, return flow, base flow, and soil moisture in the vadose zone over a certain depth cannot be taken into account in the WB method (Bhanja et al., 2018). With advanced geophysical measurements and availability of remote sensing data, the water budget method is used in hydrological modelling (Healy and Scanlon, 2010). Observations of groundwater level changes and absence of flow path preference is an advantage of using the WTF method, but it cannot separate the inter-aquifer flow (Healy and Cook, 2002). Notwithstanding the limitations, the WTF method is a widely used method for recharge estimation (Healy and Cook, 2002). Also, both WB and WTF methods showed similar recharge rates in the Indo-Gangetic basin (Bhanja et al., 2018). The advantage of the DWTF technique is the estimation of specific yield at the watershed scale which is well suited for fractured aquifers where a high density of abandoned boreholes can be used for water level measurements (Maréchal et al., 2018). While precipitation remains the main source of recharge, canal seepage (Dochartaigh et al., 2019) and return flow from irrigation can also be significant contributors of groundwater recharge (Joshi et al., 2018; Maréchal et al., 2006).

Groundwater recharge models are categorized into four classes, i.e. empirical models (precipitation-recharge), soil water balance models, coupled soil water balance hydrological models, and Richards equation-based models (Moeck et al., 2016). Groundwater recharge from models is uncertain due to model parameter uncertainty, parameterization schemes, soil heterogeneity, climate forcing data resolution, the lag time between precipitation and recharge (Xie et al., 2018), unconstrained feedbacks between recharge and climate, and groundwater pumping and irrigation practices in the agricultural areas (Meixner et al., 2016; Steward et al., 2013). There is a time lag in the groundwater recharge and the monsoon rainfall; therefore, groundwater storage anomalies are compared for a water year to capture the response of groundwater storage change to precipitation variability (Asoka et al., 2018). Inclusion of soil moisture and evapotranspiration can help in constraining estimation of groundwater recharge modelling uncertainty (Xie et al., 2018). Multi-model ensemble having different model structure and parameterization schemes to estimate groundwater recharge provides a more reliable estimate and minimizes the uncertainty (Zhu et al., 2020). Proper integration of groundwater flow paths and recharge schemes in models and response of changes in frequency and intensity of precipitation will reduce the uncertainty in recharge estimation (Meixner et al., 2016).

Overall, we find that there are challenges associated with understanding the aquifer characteristics required for groundwater recharge estimates. For instance, alluvial plains of northern and eastern parts of India have high permeability leading to higher recharge rates. In contrast, the central and southern parts have fractured crystalline cratonic rocks leading to low permeability and comparatively lower recharge rates (Asoka et al., 2017; Bhanja et al., 2019b; MacDonald et al., 2016). Inclusion of DWTF in the decision-support framework for groundwater management with 3-D map of specific yield for crystalline rock aquifers can be used as a tool for groundwater management at the basin scale (Dewandel et al., 2017, 2010; Maréchal et al., 2018). There exist considerable challenges in the estimation of groundwater recharge in India due to uncertainty in the recharge estimation approaches based on in-situ data and hydrological models. These uncertainties need to be quantified appropriately, especially in the regions that are facing severe groundwater depletion challenges. Observations of aquifer properties and appropriate spatial and temporal resolutions of monitoring wells are needed. Moreover, the influence of groundwater pumping on recharge can be estimated using the observations from deep aquifers, which unfortunately do not exist in sufficient numbers in India.

5.4. Impacts of observed and projected climate change

Evaluating the response of groundwater to climate change is essential for the planning and sustainable management of groundwater resources in India. Climate variability and climate change influence groundwater storage variability (Aeschbach-Hertig and Gleeson, 2012; Alley et al., 2002; Döll, 2009; Green et al., 2011; Konikow and Kendy, 2005; Sophocleous, 2002; Taylor et al., 2013). India has experienced significant changes in precipitation and air temperature during 1951–2016 (Fig. 8), which has influenced groundwater storage variability. For instance, a large part of the Indo-Gangetic Plain, northeast India, Western Ghats, and parts of central India have experienced a decline in the rainfall (Fig. 8b). The temperature has significantly (P-value < 0.05) increased (~0.8° C) in the western, northeastern, and peninsular India (Fig. 8d). Moreover, there is considerable spatial and temporal variability in the changes in the monsoon season rainfall in India (Fig. 9). For instance, the decline in the monsoon season precipitation in central India during 1951–2018 has been largely contributed by the decline in the late monsoon (August-September) season

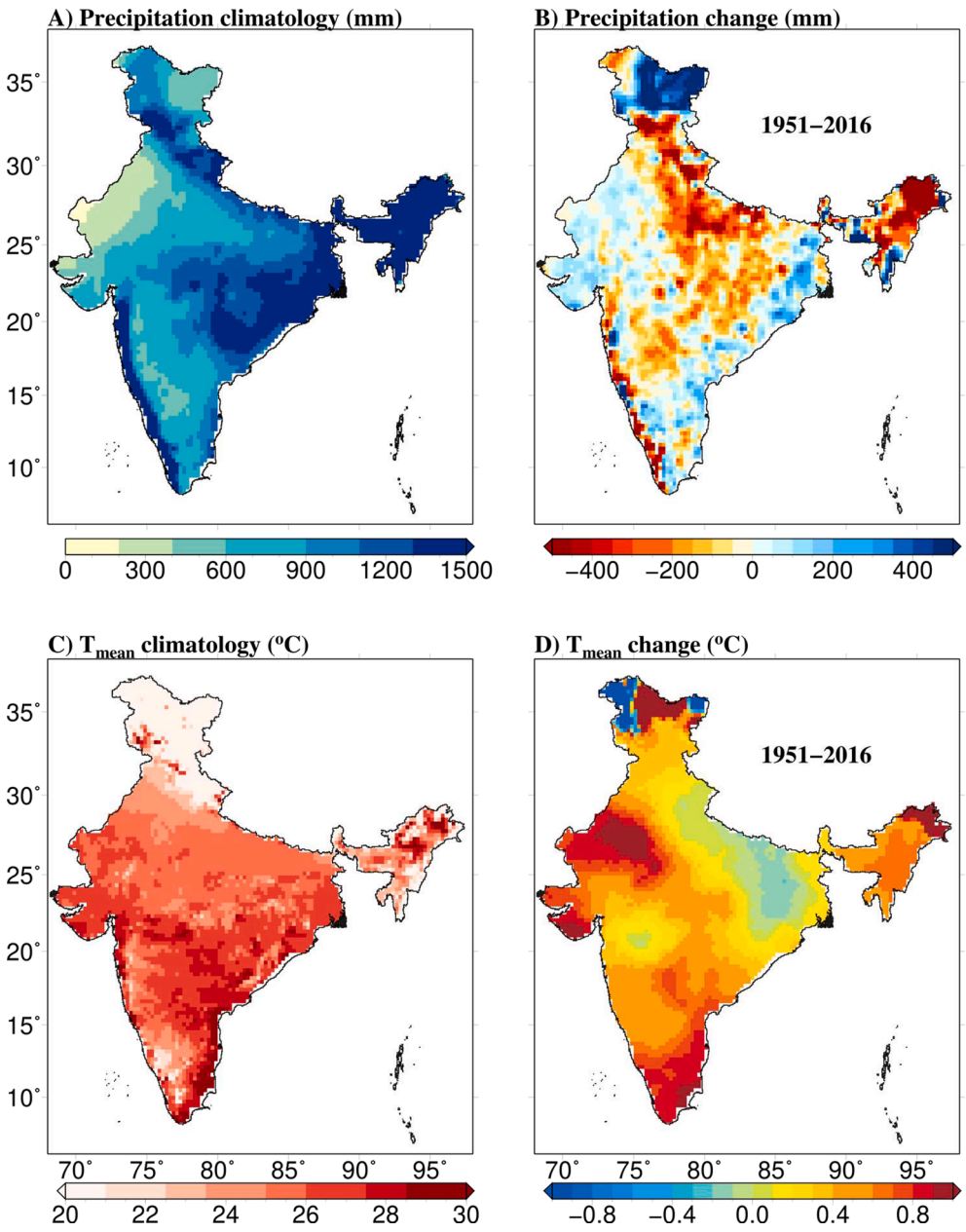


Fig. 8. The climatology and change in precipitation and temperature at 0.25° resolution during 1951–2016. A) The climatology of annual precipitation (mm) B) The change in annual precipitation C) The climatology of average annual mean temperature (°C) and D) The change in annual mean temperature. The gridded precipitation and temperature data are obtained from Indian Meteorological Department (IMD). The change (trend multiplied by number of years) in precipitation and temperature was estimated using Mann–Kendall trend test and Sen's slope method.

precipitation (Fig. 9). Precipitation in the late monsoon season in central India has declined by 18.6% (P-value < 0.05) while it has increased in western India during the early monsoon season (June-July, Fig. 9). Not only changes in the mean climate were observed in India during 1951–2016, but a significant change has also occurred in the rainfall characteristics. For instance, Asoka et al. (2018) reported that the number of rainy days with low-intensity precipitation has declined while the extreme precipitation events have increased in India. Changes in mean and extreme climate have implications for groundwater storage variability in India (Asoka et al., 2017; Bhanja et al., 2018; Meghwal et al., 2019).

Groundwater has less sensitivity to rainfall variations annually, or inter annually than surface water (Giordano, 2009), however, there exists a strong connection between groundwater and the monsoon season precipitation (Asoka et al., 2017; Bhanja et al., 2018; Rodell et al., 2018; Taylor et al., 2013). Positive or negative median trends from 1996 to 2013 can be observed in well recharge with an increase or decline of precipitation, respectively (Asoka et al., 2018). For instance, low-intensity precipitation favours the monsoon season groundwater recharge in north India, while high-intensity precipitation is more useful for groundwater recharge in South India (Asoka et al., 2018) [Fig. 10]. Bhanja et al. (2019b) also showed that precipitation played a significant role in groundwater recharge in six Indian river basins (e.g., Mahanadi, Sabarmati, Narmada, Tapi), which are not intensively irrigated with withdrawal rate less than 75 mm a year.

Understanding of the projected future changes in groundwater storage in India is essential. However, it is not straight forward. There are uncertainties associated with groundwater recharge under climate change (Holman, 2006), which are linked to precipitation projections under the future climate (Chaturvedi et al., 2012; Kumar et al., 2013; Niraula et al., 2017; Turner and Slingo, 2009). A majority of the global climate models projected an increase in the Indian summer monsoon precipitation (Chaturvedi et al., 2012; Menon et al., 2013). However, most of the global climate models do not show the desired skills to simulate the critical features of the summer monsoon variability (Ashfaq et al., 2017). The majority of CMIP5 climate models missed the observed

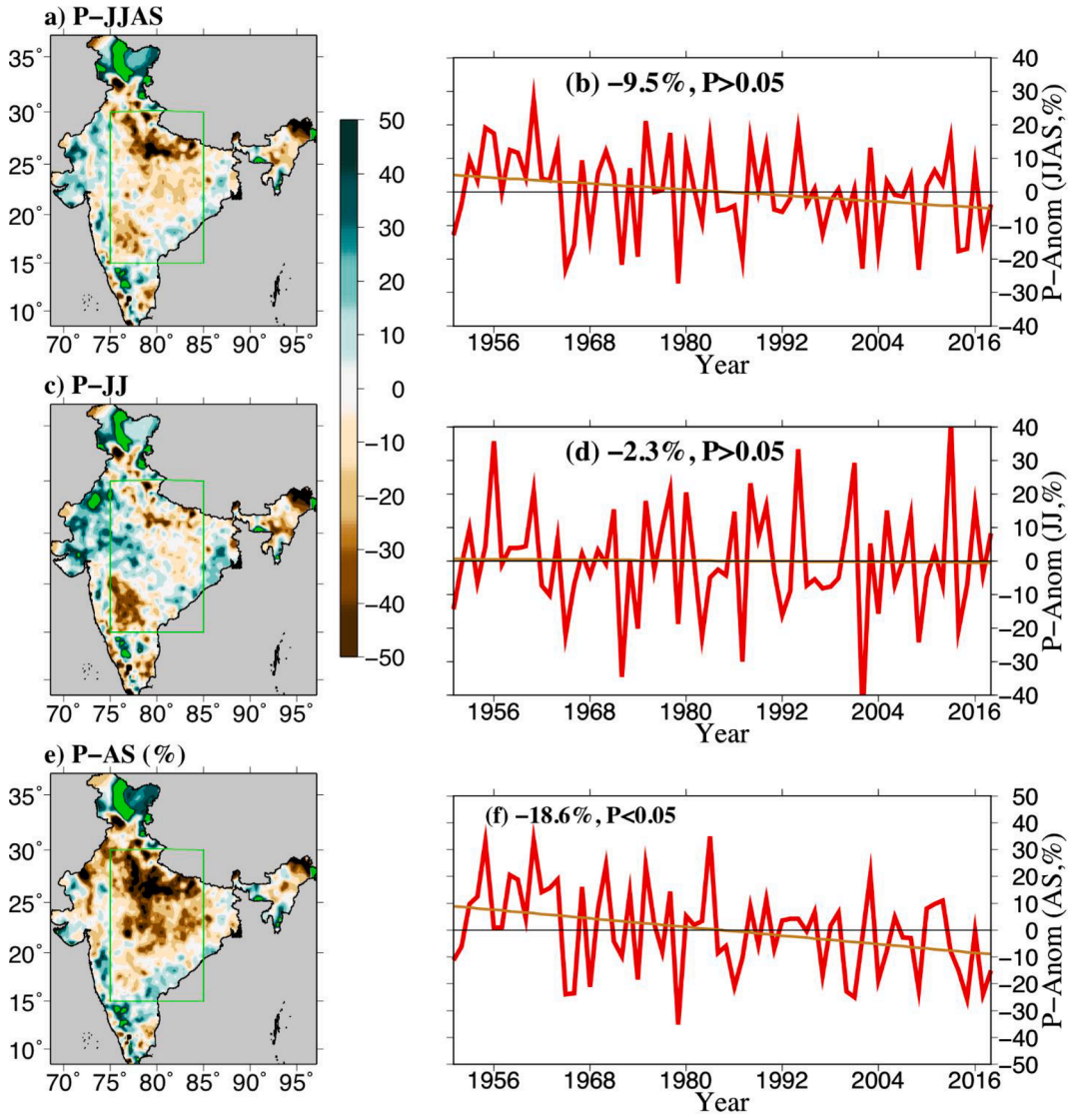


Fig. 9. (a) Changes in the monsoon season (June-September) precipitation during 1951–2018, (b) changes in the monsoon season precipitation in central India, (c) changes in precipitation during the early monsoon season (June-July), (d) same as (b) but for the early monsoon season, (e) changes in precipitation during the late monsoon (August-September), and (f) same as (b) but for the late monsoon season. Changes were estimated using Mann-Kendall trend test and Sen's slope method. Green box in (a, c, e) shows central India region. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(1950–2005) Indian summer monsoon rainfall declining trends (Saha et al., 2014). Aadhar and Mishra (2019) used the projections from the six global climate models that show reasonable skills over India and reported a substantial rise in dryness in South Asia under the warming climate. Mishra et al. (2020) reported that the concurrent hot and dry monsoon extremes would increase in India under the future climate. Similarly, the frequency of severe droughts is also projected to increase in India (Aadhar and Mishra, 2020). Wu et al. (2020) reported that groundwater storage is projected to increase in northwestern India under the future climate in the absence of groundwater abstraction for irrigation primarily due to projected increase in precipitation. Given the uncertainty in the monsoon season precipitation projections (Aadhar and Mishra, 2020, 2019; Menon et al., 2013), approaches based on the sensitivity analysis are required to understand the changes in groundwater storage in India in the warming climate. Projected changes in precipitation will affect groundwater recharge, while changes in the atmospheric water demands can influence groundwater abstraction (Taylor et al., 2013). Longer droughts and projected increase in irrigation water demands can further pose challenges to already depleting groundwater resources in India.

In addition to changes in the mean and extreme climate, projected changes in groundwater storage will depend on the changes in the irrigated areas and irrigation demands (Döll et al., 2016; Fischer et al., 2007; Wada et al., 2013; Yoshikawa et al., 2014), changes in irrigation efficiency and cropping systems (Fishman et al. (2015); Foley et al. (2011); Jägermeyr et al. (2015, 2016); Lin Lawell (2016); Pfeiffer and Lin (2014); Scott et al. (2014)). Groundwater projections do not consider the increase in irrigated areas and water demand, which may be mitigated substantially with the improvement in irrigation efficiency (Wada et al., 2014). Uncertainties in the emission scenarios in global climate models, downscaling, hydrologic models, climate change impact on land cover change make predicting climate impacts on groundwater a challenging task (Döll, 2009). Groundwater component representation in global climate models (GCMs) needs to be improved by including complex interactions between humans and the climate (Taylor et al., 2013) and proper representation of land cover change, water withdrawals data (Kurylyk and MacQuarrie, 2013). Also, for future recharge scenarios, the model ensemble of different GCMs with different

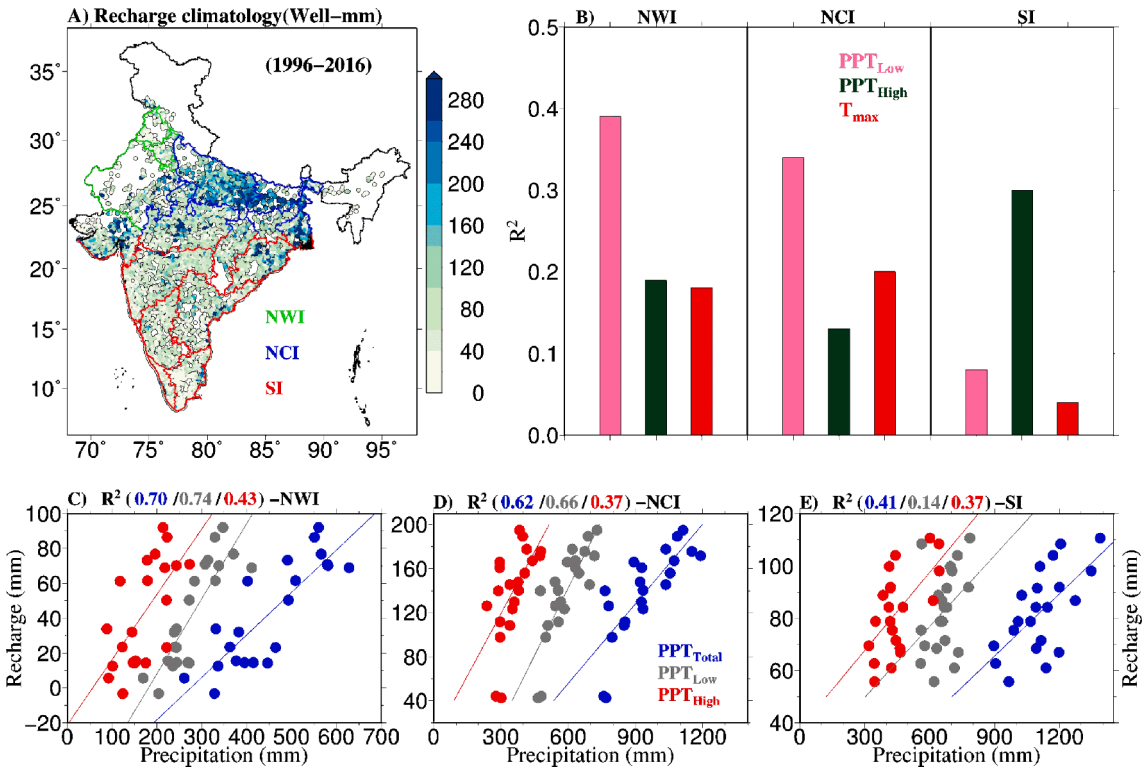


Fig. 10. A) Monsoon season groundwater recharge estimated using water table fluctuation method. B) The relative importance of low-intensity precipitation (PPT_{Low}), high-intensity precipitation (PPT_{High}), and maximum temperature (Tmax) for groundwater recharge for the northwest (NWI), north-central (NCI), and South India (SI). Relationship between groundwater recharge (mm) and precipitation amounts due to total (PPT_{Total}; blue), low (PPT_{Low}; gray), and high (PPT_{High}; red) intensity precipitation in northwest India, NWI (C), north central India, NCI (D), and South India, SI (E). Coefficient of determination (R^2) of the regression relationship for PPT_{Total}, PPT_{Low}, and PPT_{High} are given in blue, gray, and red colors. [Modified from Asoka et al., (2018)] (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

scenarios should be used in hydrological models. Therefore, understanding of the projected changes in groundwater storage in India is vital yet incomplete.

6. Synthesis, knowledge gaps, and open questions

India has witnessed rapid groundwater depletion in the last few decades, which is caused by the both natural and human factors. Climate variability and climate change have affected groundwater recharge and abstraction, causing a rapid loss in groundwater storage. For instance, a large part of the country has witnessed a decline in the monsoon season precipitation during 1951–2018. Also, the climate has become warmer by one °C during the last six decades in India. There have been noticeable changes in the characteristics of the monsoon season precipitation (Asoka et al., 2018; Singh et al., 2014). The number of rainy days and rainy days with low-intensity precipitation have significantly declined in India, affecting groundwater recharge during the monsoon season, which remains the key source of supply to groundwater storage. On the other hand, the number of days with high-intensity precipitation has increased in India during the 1951–2018 period (Asoka et al., 2018), which is favourable to groundwater recharge in south India dominated by bedrock aquifers. A majority of the observation wells in India is located in shallow aquifers, and they do not provide the recharge estimates in the confined aquifers. Groundwater pumping for irrigation mostly happens from the deep aquifers for which observations for long-term are not available. Therefore, groundwater observations mostly do not provide the information of recharge and abstraction; rather, they indicate the role of climate variability on the shallow aquifers.

The GRACE satellites have provided, and GRACE-FO will continue to provide valuable information on both shallow and deep groundwater storage. The GRACE data have been beneficial for understanding the causes of groundwater depletion in India. However, relatively shorter records and other uncertainties (Long et al., 2016) associated with surface water storage from the land surface models produce challenges for understanding the role of climate and humans on groundwater systems. Land surface hydrological models fill a significant gap that exists due to the lack of observations of hydrological variables (soil moisture, evapotranspiration, and terrestrial water storage). However, the hydrological models have known uncertainties due to parameterization, process representation, and availability of input data. In addition, a majority of the models do not represent in a sufficient manner the complex processes associated with human-climate interaction that affect groundwater.

Groundwater has significantly depleted primarily due to unsustainable groundwater pumping for irrigation. Groundwater depletion is likely to continue in India due to increased climate warming and enhanced food demands for the rising population. While there are uncertainties in the climate model projections in India, a majority of the models project an increase in the summer monsoon precipitation (Menon et al., 2013; Mishra et al., 2020). However, the projected increase in the monsoon season precipitation does not avoid the possibility of droughts in India (Aadhar and Mishra, 2020). Moreover, enhanced groundwater recharge due to the projected increase in rainfall can be nullified by the increased irrigation demands during the dry season because of the rise in temperature. Understanding the impacts of projected future change in climate, cropping patterns, irrigation systems, and irrigation efficiency on groundwater storage in India remains a grand challenge. Future of food and water security in India can be significantly influenced by the way coupled human and climate systems will change in the future.

Based on our review, the following are the knowledge gaps and open questions that need attention:

- Measurement of groundwater variability is the key to finding out ways for sustainable management. Despite a large number of observational wells in India, the fraction of deep wells is significantly low, which hampers our ability to understand recharge and abstraction dynamics in the deep aquifers. Therefore, there is a need to increase the observations from the deep aquifers. The other important aspect related to the in-situ measurement of well level is associated with sampling frequency. The frequency of measurements needs to be increased (at least once every month) to understand better the role of irrigation and climate on groundwater storage variability. Moreover, the optimality of the groundwater monitoring network in different regions needs to be carefully examined.
- GRACE data provide valuable information on groundwater storage variability. However, uncertainties associated with different products based on the GRACE satellites need to be quantified. In addition, how the errors in the GRACE based observations vary spatially in the country (in different aquifer types) need to be understood. Groundwater storage estimates based on the GRACE can have uncertainties based on soil moisture from the land surface models. Therefore, there is a need to properly evaluate the land surface models against the in-situ observations of soil moisture, evapotranspiration, groundwater level, and streamflow in India.
- Land surface hydrological models provide essential information that helps us in understanding the variability in groundwater storage. However, current generation land surface models lack the representation of complex processes associated with human-climate interactions. For instance, agricultural and irrigation practices have considerable heterogeneity, which is often not represented in the physical processes in the models. There is a strong need to develop new or enhance the existing land surface hydrological models for the Indian condition to improve our understanding related to drivers of groundwater storage variability in India. Since models require observations for input or calibration and evaluation, observational networks and testbeds need to be established in India.
- New approaches of downscaling the groundwater observations at different spatial and temporal resolution are required for better understanding of groundwater storage variability in response to climate variability and anthropogenic intervention. Machine learning-based approach combined with process-based models needs to be evaluated in different regions of the country.
- While the importance of groundwater for food and freshwater security in India has been broadly studied, there is little knowledge about how the significance of groundwater will change in the future in India. Therefore, there is a need to develop realistic projections of population, food demands, land-use change, irrigated area, irrigation technology, and irrigation efficiency for India. Moreover, how these factors will be influenced by climate change need to be understood to some extent to develop future scenarios of groundwater in India.
- The sensitivity of groundwater recharge to changes in precipitation and air temperature needs to be quantified for different aquifers in the country. Moreover, the influence of extreme events (droughts and floods) on groundwater recharge during the monsoon season and abstraction during the dry season is not well understood.
- The role of groundwater conservation and enhancing groundwater recharge in the observed and projected future climate needs to be quantified, given the other uncertainties. Moreover, the future hotspots of groundwater depletion and its drivers need to be identified to develop a framework for food and water security in those regions.
- The problem of groundwater depletion in India is challenging. An integrated framework of in-situ and satellite-based observations, process models, agricultural and irrigation practices, and political and stakeholders' participation is required to provide sustainable solutions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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